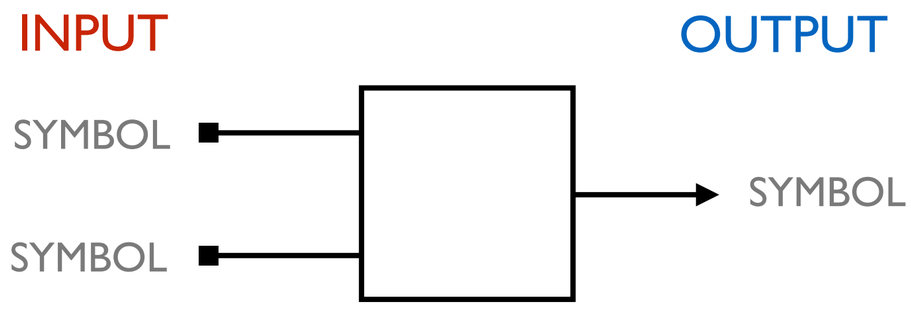
# 3. Logic Gates & Combinatorial Circuits

## Logic Gates

**Logic gates** are the basic building blocks of most human-made computers. By combining logic gates into complex networks, we can ultimately achieve complete computational power.

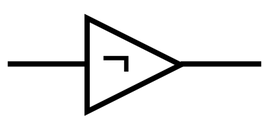
A single logic gate is actually a very, very simple computer. It takes one or two symbols as input (depending on the gate), and produces a single symbol as output.

There are billions of logic gates in a normal desktop processor.



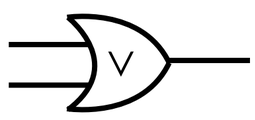

We'll use three basic types of logic gates. Each is defined by its input-output profile:

NOT-GATE



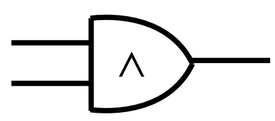

| IN | OUT |
|---|---|
| 0 | 1 |
| 1 | 0 |

OR-GATE




| IN 1 | IN 2 | OUT |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

AND-GATE




| IN 1 | IN 2 | OUT |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

To understand logic gates, don't think of the input and output symbols as actual "1"s and "0"s. They are abstract values that are defined simply as opposites.

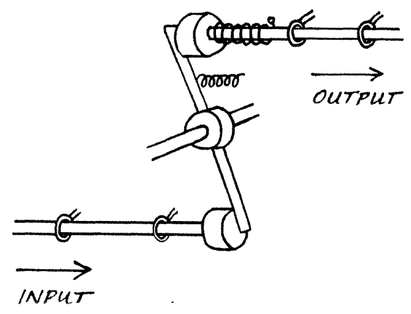
The logic gate itself is a kind of abstract machine which matches inputs with outputs.

| IN | OUT |
|---|---|
| 0 | 1 |
| 1 | 0 |

| IN | OUT |
|---|---|
| off | on |
| on | off |

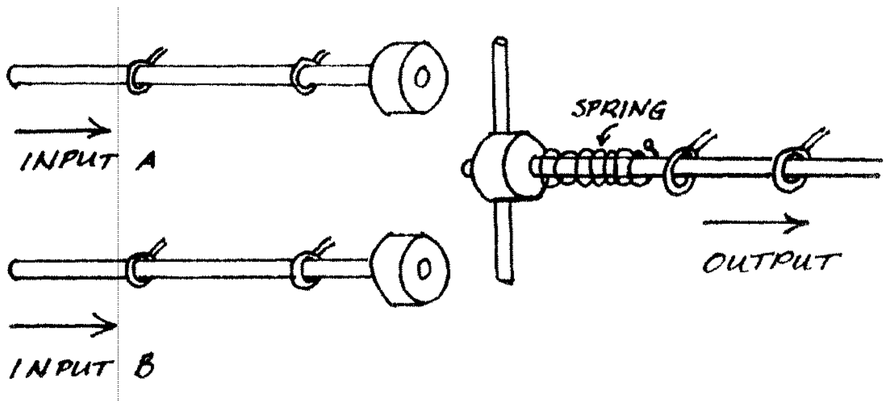
Thought of as abstract machines, logic gates work "instantaneously" and never make mistakes or wear out.

Physical implementations of logic gates take time to process, make mistakes, and are subject to entropy.

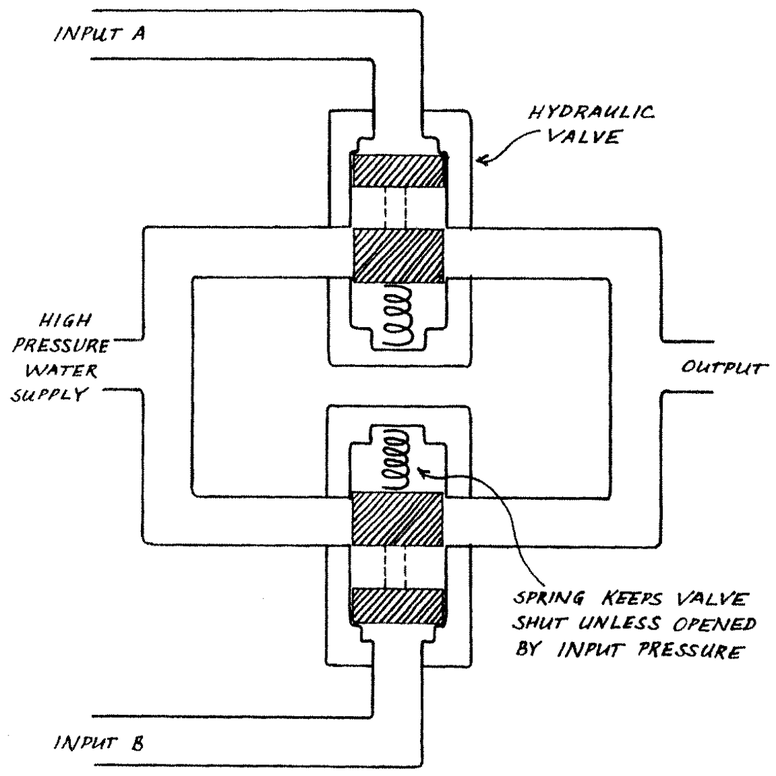


# From Physics to Logic

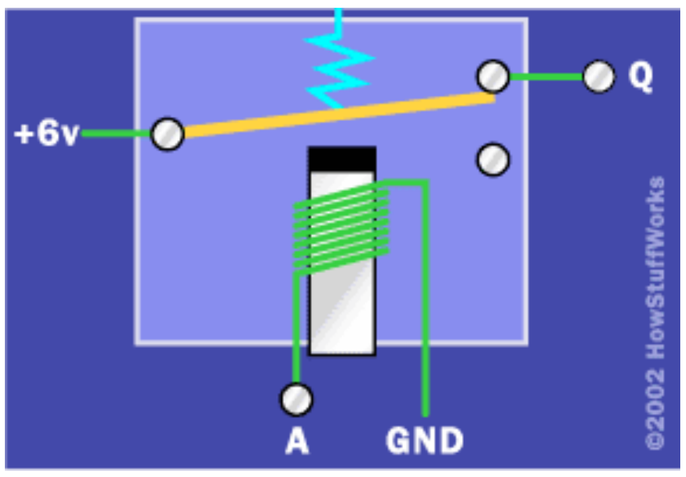
Logic gates can be implemented using any number of different physical designs and physical materials.



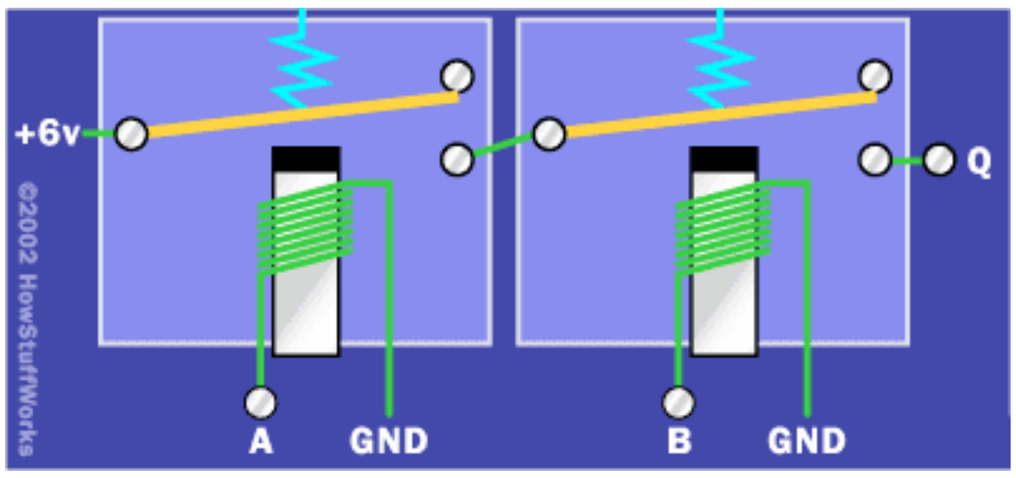

Physical implementation of an OR-gate with a stick and spring mechanism.




Physical implementation of an OR-gate with hydraulic valves.




Physical implementation of a NOT-gate with electromagnets.




Physical implementation of an AND-gate with electromagnets.

Logic gates themselves can be thought of as abstractions from concrete machines.

They abstract away from the internal description of a machine by specifying only inputs and outputs.

And they abstract away from the external description by specifying inputs and outputs only as “0” and “1”.

| IN | OUT |
|---|---|
| 0 | 1 |
| 1 | 0 |

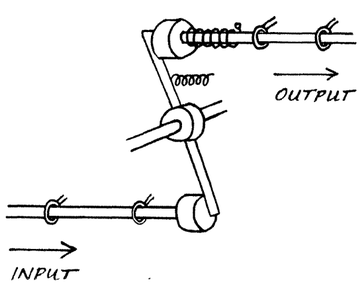
A NOT-gate.

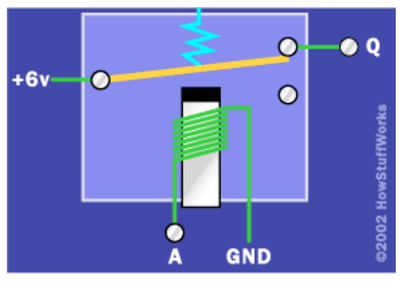
Abstraction over external description.

| IN | OUT |
|---|---|
| → | ← |
| ← | → |

| IN | OUT |
|---|---|
| ⚡ | ⚡ |
| ⚡ | ⚡ |

Abstraction over internal description.

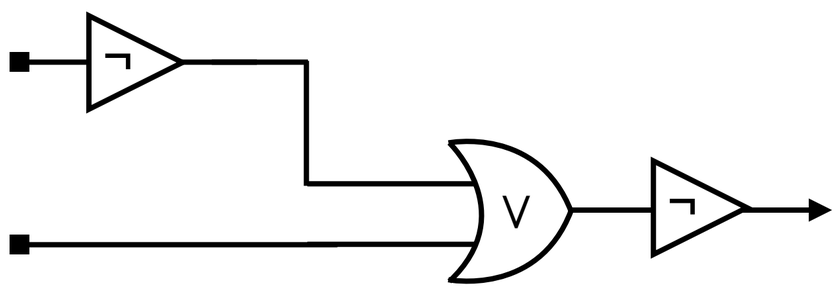



Physical implementations of a NOT-gate.

## Combinatorial Circuits: Concept

Logic gates can be combined together into a circuit, called a **combinatorial circuit** or **CC.** Here's an example:




Combinatorial circuits are made up of **logic gates** connected by **links.** The links represent pathways that symbols may flow through.

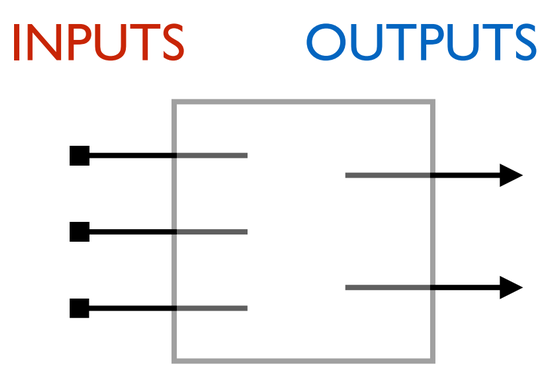
In a combinatorial circuit, we abstract from time. Symbols pass "instantaneously" from inputs, through the network of links and logic gates, to the final output.

**Input/output control.** In a factory-made computer, the inputs are determined by the user, while the outputs are determined by the machine (given the input). In a "natural" computer, like the brain of an animal, the inputs are determined by the natural environment, but the outputs are still determined by the machine.

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## Combinatorial Circuits: Definition

**Input and output.** A combinatorial circuit can have any number of inputs and any number of outputs.

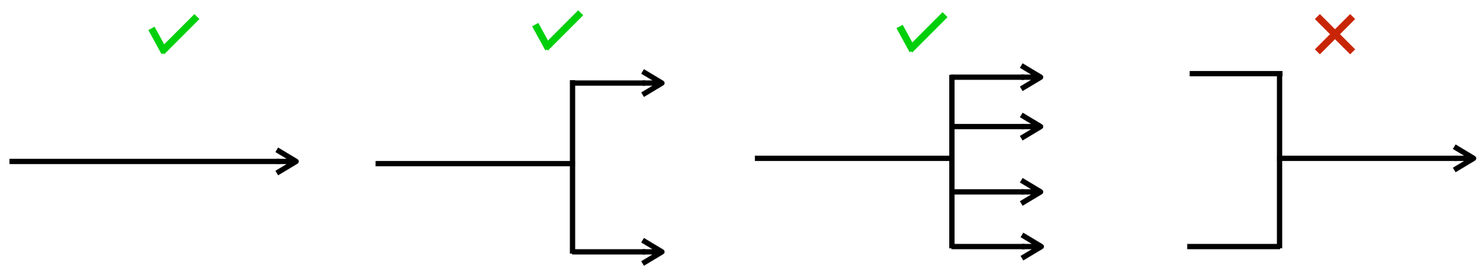



For clarity, mark links which are connected to the input with small squares. Mark links connected to the output with small arrows.

Each input leads either into a logic gate or directly to an output.

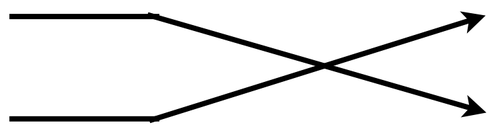
---

**Links.** Logic gates are connected to one another by **links.** Links may split any number of times. When a link splits, it carries the same symbol to both of its branches. Links may split, but they can't merge, because how would you combine a 0 and a 1?

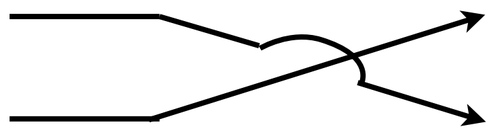



When drawing your circuit diagrams, the link-*lines* may sometimes cross. This is OK, it doesn't say anything about the circuit itself, which is an abstract network of logic gates. Line crossing is just a side effect of drawing in two dimensions.

So the following diagram is fine:



But for visual clarity, introduce a "hop":

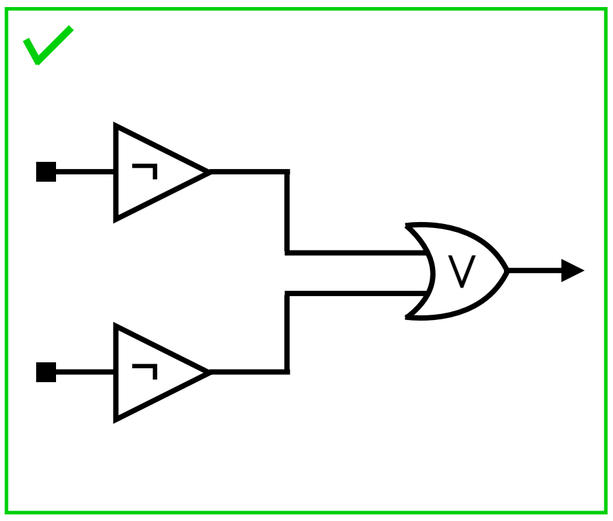


**CC Definition.** A combinatorial circuit can now be officially defined as follows:

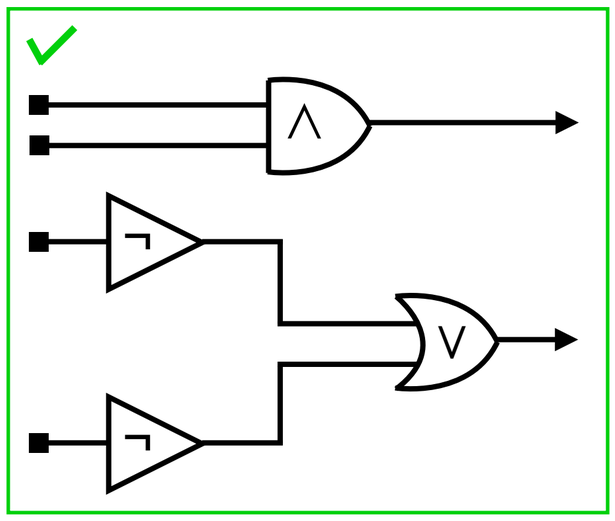
A **combinatorial circuit** is any network of logic gates and links where:

- there are no loops;
- there are no merged links;
- every free end is either an input or output.

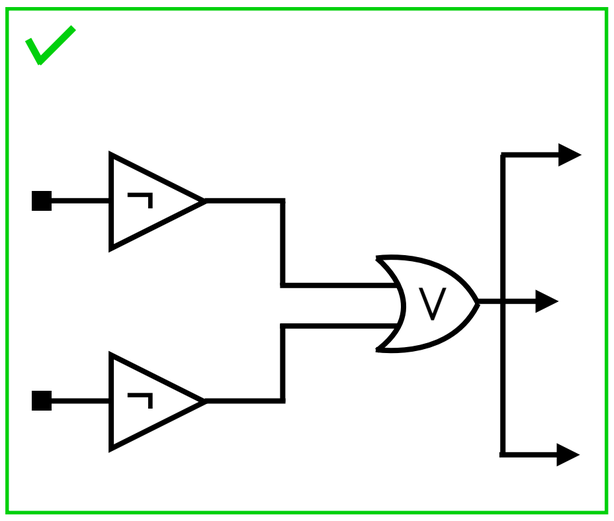
A standard case.



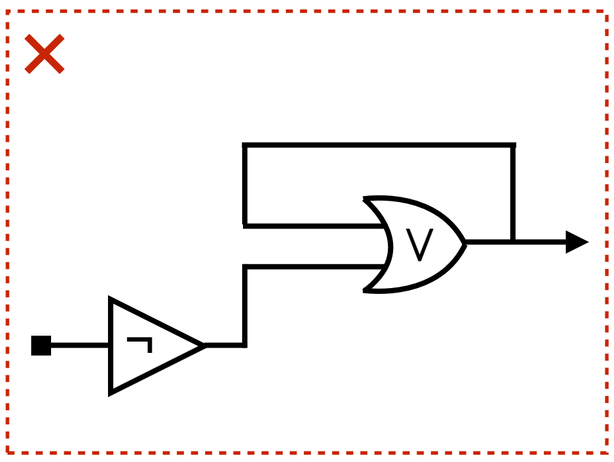
There may be disconnected parts.



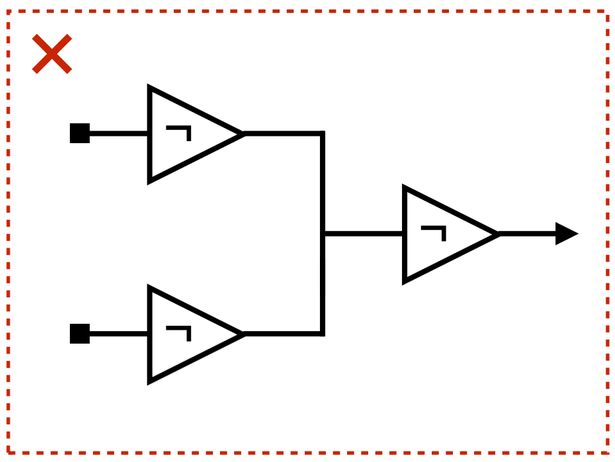
There may be more outputs than inputs.



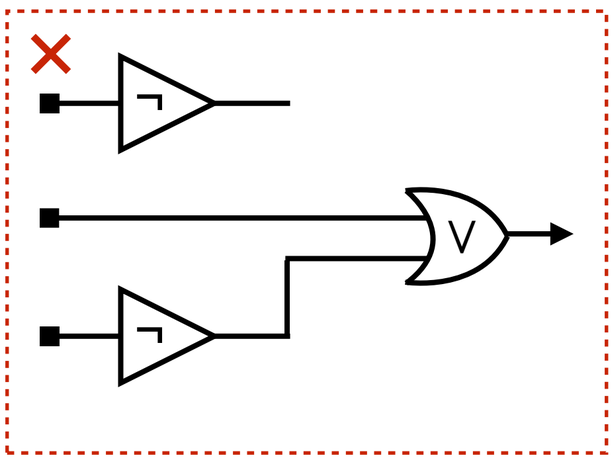
No loops allowed.



No merged links allowed.

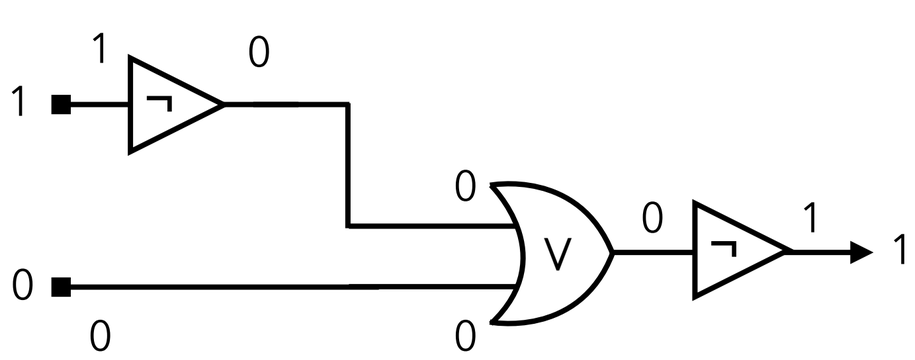


No free ends (except in/out).

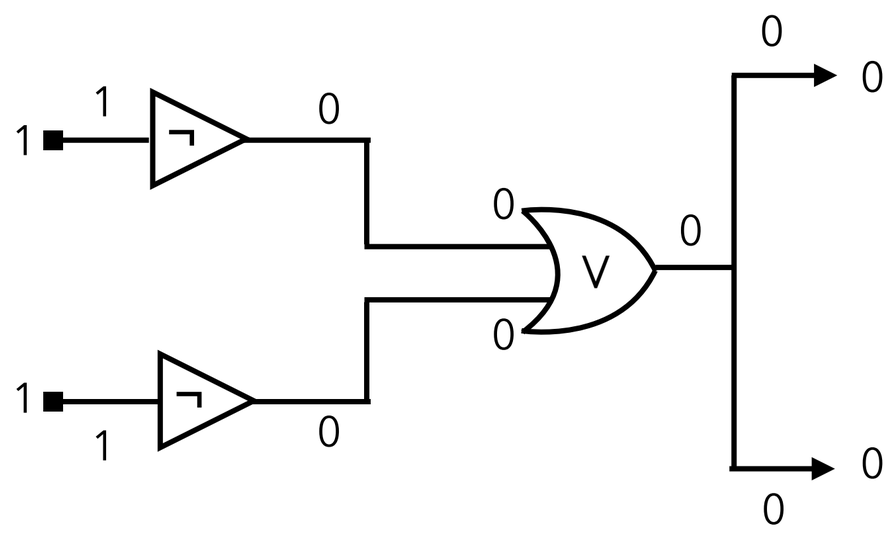


## Combinatorial Circuits: Flow of Information

Here are two sample circuits and their input-output tables. Each circuit is labeled with one possible set of inputs, as well as their values as they propagate through the circuit.




| IN 1 | IN 2 | OUT |
|---|---|---|
| | | |
| | | |
| 1 | 0 | 1 |
| 1 | 1 | 0 |




| IN 1 | IN 2 | OUT 1 | OUT 2 |
|---|---|---|---|
| | | | |
| | | | |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |

## In-Class Problems

Construct a combinatorial circuit for each input-output table:

1. Not

| IN 1 | OUT |
|---|---|
| 0 | 1 |
| 1 | 0 |

2. Identity

| IN 1 | OUT |
|---|---|
| 0 | 0 |
| 1 | 1 |

3. And

| IN 1 | IN 2 | OUT |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

4. Nand

| IN 1 | IN 2 | OUT |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

5. Or

| IN 1 | IN 2 | OUT |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

6. Nor

| IN 1 | IN 2 | OUT |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

7. All-1

| IN 1 | OUT |
|---|---|
| 0 | 1 |
| 1 | 1 |

8. All-0

| IN 1 | OUT |
|---|---|
| 0 | 0 |
| 1 | 0 |

9. Split

| IN 1 | IN 2 | OUT 1 | OUT 2 |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |